

Nicole Sandra-Yaffa Dumont

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Education

Ph.D in Computer Science

Diploma in Theoretical Neuroscience

University of Waterloo

📅 Sept 2019 – April 2025

📍 Waterloo, Canada

- Dissertation title: *Symbols, Dynamics, and Maps: A Neurosymbolic Approach to Spatial Cognition*
- Supervised by Dr. Chris Eliasmith and Dr. Jeff Orchard
- Neuro-Symbolic Modeling for Spatial Representation & Mapping
 - Developed Spatial Semantic Pointers (SSPs) as a grid cell-based representation of continuous features. SSPs enable structured, hierarchical encoding across multiple domains (e.g., spatial, temporal, conceptual).
 - Built spiking neural network models for path integration, cognitive mapping, and SLAM that use neurosymbolic methods to bind continuous and discrete features.
- Reinforcement Learning & Action Selection
 - Designed biologically plausible online learning rules for RL in continuous-time settings using State Space Model-based memory systems.
 - Improved sample efficiency during navigation using SSPs and structured scene representations as inputs to deep RL frameworks.
 - Integrated SSPs into Bayesian Optimization frameworks for active exploration, achieving up to 175x reduction in compute time in multi-agent environments.

Computational Mathematics (Masters of Mathematics – Co-op Program)

University of Waterloo

📅 Sept 2017 – April 2019

📍 Waterloo, Canada

- Developed a method for robust optimization of an asset pricing model applied to carbon emissions.

Honours Mathematics and Physics (Bachelors of Science)

McMaster University

📅 Sept 2012 – April 2017

📍 Hamilton, Canada

Experience

Neuromorphic Scientist

Applied Brain Research

📅 Jan 2023 – Jan 2024

📍 Waterloo, Canada

- Integrated a Spatial Semantic Pointer-based SLAM algorithm in a MuJoCo simulation environment and deployed the algorithm on neuromorphic hardware (Intel's Loihi 1 chip).

Research Associate

Cayuga Research

📅 May 2018 – Aug 2019

📍 Waterloo, Canada

- Consulting work focused on the development and implementation of optimization methods and data driven solutions to industrial problems.

Research Assistant

Ayers Research Group, McMaster University

📅 May 2015 – Aug 2015

📍 Hamilton, Canada

- Developed a semi-definite optimization algorithm to constrain a density matrix for representing a many-electron quantum system.

Research

Publications & Conferences

† = These authors contributed equally

- Furlong, P. M., Simone, K., **Dumont, N.S.**, Bartlett, M., Stewart, T. C., Orchard, J., & Eliasmith, C. (2024). Biologically-plausible markov chain monte carlo sampling from vector symbolic algebra-encoded distributions. In *ICANN*. Springer [\[Link\]](#)

- Furlong, P. M., **Dumont, N.S.**, & Orchard, J. (2024). A recurrent dynamic model for efficient bayesian optimization. In *NICE*. IEEE [\[Link\]](#)
- **Dumont, N.S.**, Furlong, P. M., Orchard, J., & Eliasmith, C. (2023). Exploiting semantic information in a spiking neural slam system. *Frontiers in Neuroscience*, 17 [\[Link\]](#)
- **Dumont, N.S.**[†], Stöckel, A.[†], Furlong, P. M.[†], Bartlett, M., Eliasmith, C., & Stewart, T. C. (2023). Biologically-based computation: How neural details and dynamics are suited for implementing a variety of algorithms. *Brain Sciences*, 13(2) [\[Link\]](#)
- Bartlett, M.[†], Simone, K.[†], **Dumont, N.S.**[†], Eliasmith, C., & Orchard, J. (2023). Improving reinforcement learning with biologically motivated continuous state representations. In *ICCM* [\[Link\]](#)
- **Dumont, N.S.**, Orchard, J., & Eliasmith, C. (2022). A model of path integration that connects neural and symbolic representation. In *CogSci* (Vol. 44, 44), Cognitive Science Society [\[Link\]](#)
- Bartlett, M., **Dumont, N.S.**[†], Furlong, M. P., & Stewart, T. C. (2022). Biologically-plausible memory for continuous time reinforcement learning. In *ICCM* [\[Link\]](#)
- Voelker, A. R., Blouw, P., Choo, X., **Dumont, N.S.**, Stewart, T. C., & Eliasmith, C. (2021). Simulating and predicting dynamical systems with spatial semantic pointers. *Neural Computation*, 33(8) [\[Link\]](#)
- **Dumont, N.S.**, Stewart, T. C., & Eliasmith, C. (2021). Spiking neural network model of simultaneous localization and mapping with spatial semantic pointers. In *Cosyne* [\[Link\]](#)
- Coleman, T. F., **Dumont, N.S.**, Li, W., Liu, W., & Rubtsov, A. (2021). Optimal pricing of climate risk. *Computational Economics* [\[Link\]](#)
- **Dumont, N.S.**, & Eliasmith, C. (2020). Accurate representation for spatial cognition using grid cells. In *CogSci*, Cognitive Science Society [\[Link\]](#)

Manuscripts Under Review

- Furlong, P.M.[†], **Dumont, N.S.**[†], Antonova, R.[†], Orchard, J., & Eliasmith, C. (In Revision). Compositional neurosymbolic representations enable efficient active exploration. *Nature Communications*

Thesis

- **Dumont, N.S.** (2025). *Symbols, dynamics, and maps: A neurosymbolic approach to spatial cognition* (PhD Thesis, University of Waterloo, Waterloo, ON) [\[Link\]](#)

Teaching and Supervision

- **Introduction to Computational Mathematics (2022)**: I was the sole instructor for two sections of this course, teaching over 120 students. I planned the course content, and created lectures, assignments, and tests covering a range of numerical methods.
- **Logic and Computation (2019, 2020)**: I was a graduate Teaching Assistant and Instructional Apprentice for this introductory course on mathematical logic. I prepared and led tutorials, graded assignments, and assisted students during office hours.
- **Student supervision (2020)**: I supervised an undergraduate research assistant on a project developing a novel method for learning successor features for RL.

Academic Service

- **2025** – Co-organizer of the Language and Thought Topic Area, Telluride Neuromorphic Cognition Engineering Workshop
- **2024** – Invited speaker, Telluride Neuromorphic Cognition Engineering Workshop
- **2024, 2023** – Instructor, Nengo Summer School
- **2025** – Reviewer, Nature Communications
- **2025** – Reviewer, CogSci 2025
- **2024** – Reviewer, Frontiers in Neuroscience

Scholarships & Awards

- **2024** – ENNS Best Paper Award
- **2022** – Barbara Hayes-Roth Award for Women in Math and Computer Science
- **2020, 2019** – Go-Bell Scholarship
- **2019** – Provost Doctoral Entrance Award for Women
- **2019** – Keith & Debbie Geddes Graduate Scholarship
- **2014** – The Emanuel Williams Scholarship in Physics